

2. Construct a function of the form $y = \int_a^x f(t) dt$ if $\frac{dy}{dx} = \ln(\sin x + 5)$, and $y = 3$ when $x = 2$ $\frac{1}{(2)}$

$$y = \int_2^x (\ln(\sin t + 5)) dt + 3$$

check notes of T.B.

3. (1 marks)

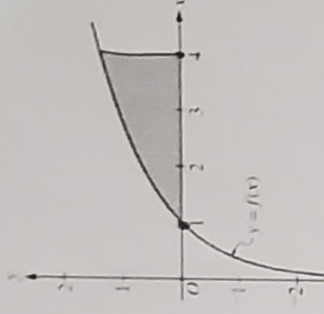
The function f is given by $f(x) = \ln x$. The graph of f is shown above. Which of the following limits is equal to the area of the shaded region?

(A) $\lim_{n \rightarrow \infty} \sum_{k=1}^n \left(1 + \ln \left(\frac{3k}{n} \right) \right) \frac{3}{n}$

(B) $\lim_{n \rightarrow \infty} \sum_{k=1}^n \ln \left(1 + \frac{3k}{n} \right) \frac{3}{n}$

(C) $\lim_{n \rightarrow \infty} \sum_{k=1}^n \ln \left(\frac{4}{n} \right) \left(1 + \frac{4k}{n} \right)$

(D) $\lim_{n \rightarrow \infty} \sum_{k=1}^n \ln \left(1 + \frac{4k}{n} \right) \frac{4}{n}$



Thinking (6 marks)

1. Determine (the positive) k value if $\int_0^k \frac{x}{x^2+4} dx = \frac{1}{2} \ln 4$

$$u(k) = k^2 + 4 \quad (2 \text{ marks})$$

$$\int_0^k \frac{x}{x^2+4} dx = \frac{1}{2} \ln(4) \quad \text{so find integral first}$$

$$\text{Use } u = x^2 + 4 \Rightarrow \frac{1}{2} \ln(u) \Big|_{x=0}^{x=k}$$

$$\frac{1}{2} \int_0^k \frac{1}{u} du = \frac{1}{2} \ln(u(k)) - \frac{1}{2} \ln(u(0)) = \frac{1}{2} \ln(4)$$

$$\frac{1}{2} \ln(k^2+4) = \ln(4) \quad |k = \sqrt{12}$$

$$\ln(k^2+4) = 2 \ln(4)$$

$$k^2+4 = 16 \quad \checkmark$$

$$k = \sqrt{12} \quad (2 \text{ marks})$$

2. Using properties of the integrals show that $0 \leq \int_5^{10} \frac{x^2}{x^4+x^2+1} dx \leq 0.1$

$$5 \leq x \leq 10 \quad \text{so we must find max and min of } \frac{x^2}{x^4+x^2+1}$$

$$\frac{2x}{x^4+x^2+1} \leq 0.098 \quad \text{Show function } \leq 0.1$$

$$\frac{2x}{x^4+x^2+1} \leq 0.098 \quad \text{Miner range } 1 \approx 0.099$$

$$\frac{10^2}{10^4+10^2+1} = 0.00990$$

$$\text{Max}$$

3. If f is continuous on $[0,1]$, prove that $\int_0^1 f(x) dx = \int_0^1 f(1-x) dx$

$$\text{Let } a = 1-x$$

$$\int_0^1 f(x) dx = \int_{x=0}^{x=1} f(a) da$$

since a ranges from

$0 \rightarrow 1$ smoothly and x ranges from $1 \rightarrow 0$ smoothly the integral is the same as values at mb function

$$\int_0^1 f(a) da = \int_{a=1}^{a=0} f(a) da \rightarrow - \int_{a=1}^{a=0} f(a) da = \int_0^1 f(a) da$$

Cont. on paper

1. Let $g(x) = \int_0^x f(t) dt$, where f is the function whose graph is shown.

a) Evaluate $g(0)$, $g(1)$, $g(2)$, $g(3)$, and $g(5)$.

$$g(0) = \int_0^0 f(t) dt = 0$$

$$g(1) = 2, \quad g(2) = 2, \quad g(3) = 5$$

$$g(4) = 5$$

$$g(5) = 7$$

$$g(6) = 6.5$$

b) On what interval is g decreasing?

$g'(x) = f(x)$ so $f(x)$ is deriv

$g'(x) < 0$ func

(3, 7)

c) Where does g have a maximum value?

at $x=3$ as $f(x) > 0$ changes from + to -

✓

d) Sketch the graph of g in the same grid where f is given. Explain.

(2, 1) increasing (1, 2) & increasing so gets steeper
(2, 3) starting down, tangent slope gets steeper to 0

2. Write the accumulator function $A(x)$ as a polynomial of x :

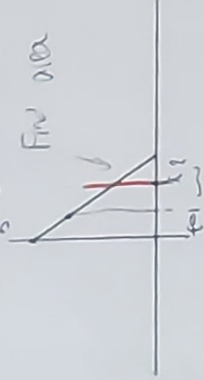
$$A(x) = \int_1^x (5-2t) dt \quad \text{for } 1 \leq t \leq 2$$

$$5-2x+3 = 8-2x$$

height

$$(x-1) \cdot \left(\frac{f(x) + f(1)}{2} \right)$$

$$A(x) = (x-1) \left(\frac{18-2x}{2} \right) \quad 1 \leq x \leq 2$$



3. Let f and g be continuous functions such that $\int_0^6 f(x) dx = 9$, $\int_3^6 f(x) dx = 5$, and $\int_0^3 g(x) dx = -7$. What is the value of $\int_0^3 \left(\frac{1}{2} f(x) - 2g(x) \right) dx$? Show work.

is the value of

$$\int_0^3 \left(\frac{1}{2} f(x) - 2g(x) \right) dx$$

$$= \frac{1}{2} \int_0^3 f(x) dx - 2 \int_0^3 g(x) dx$$

$$= \frac{1}{2} (9) - 2(-7)$$

$$= \frac{9}{2} + 14 = \frac{37}{2}$$

$$= \frac{37}{2}$$

$$= \frac{37}{2}$$

4. Find the average value of the function: $f(x) = \frac{x^2}{(1-x^3)^2}$, on $[2, 4]$.

$$\text{let } u = 1-x^3, \quad x=4 \rightarrow u=1$$

$$du = -3x^2 dx$$

$$\frac{du}{3} = -x^2 dx$$

$$\int \frac{1}{u^2} \cdot \frac{du}{3}$$

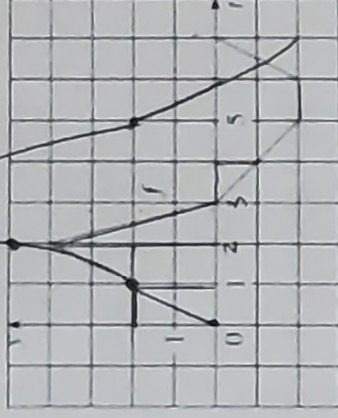
$$= \frac{1}{3} \int u^{-2} du$$

$$= \frac{1}{3} \left(-u^{-1} \right) = -\frac{1}{3u}$$

Answer 189

(-2)

(6 marks)



1/ (2 marks)

(1 mark)

$$\int_0^3 \left(\frac{1}{2} f(x) - 2g(x) \right) dx \rightarrow \frac{1}{2} (4) - 2(7)$$

$$2-14 = -12$$

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$$\int_0^3 \left(\frac{1}{2} f(x) - 2g(x) \right) dx = -12$$

5. Evaluate the integral:

a) $\int_0^1 (x^e + e^x) dx$ (1)

$$\int_0^1 x^e dx + \int_0^1 e^x dx$$

$$\frac{x^{e+1}}{e+1} \Big|_0^1 + e^x \Big|_0^1$$

$$\left(\frac{1}{e+1} - 0 \right) + (e^1 - e^0)$$

$$\frac{1}{e+1} + e - 1$$

For c) and d) find the points of discontinuity of the integrand and use area to evaluate it. (2)

c) $\int_{-2}^2 f(x) dx$ where $f(x) = \begin{cases} 2 & \text{if } -2 \leq x \leq 0 \\ 4 - x^2 & \text{if } 0 < x \leq 2 \end{cases}$

$$\int_{-2}^0 2 dx + \int_0^2 (4 - x^2) dx$$

$$2(2) + \left(4x - \frac{1}{3}x^3 \right) \Big|_0^2$$

d) $\int_{-1}^3 f(x) dx$ where $f(x) = \frac{(x-1)(x+2)}{|x-1|}$

discontinuous at $x=1$

$x > 1$ then

$$\int_{-1}^1 f(x) dx + \int_1^3 f(x) dx$$

$$-\left(\frac{x^2}{2} - 2x \right) \Big|_{-1}^1 + \left(\frac{x^2}{2} + 2x \right) \Big|_1^3 = 12$$

6. Use antiderivative part of the FTC to calculate $\frac{d}{dx} \left[\int_0^{1/\sqrt{x}} \frac{t^2}{1+t^4} dt \right]$

$$\frac{d}{dx} \left[\int_0^{1/\sqrt{x}} f(t) dt \right] = f\left(\frac{1}{\sqrt{x}}\right) \cdot \frac{d}{dx} \left(\frac{1}{\sqrt{x}} \right)$$

$$= \left(\frac{\left(\frac{1}{\sqrt{x}} \right)^2}{1 + \left(\frac{1}{\sqrt{x}} \right)^4} \right) \cdot \left(-\frac{1}{2} \cdot x^{-3/2} \right) = -\frac{1}{2} \cdot x^{-3/2} \cdot \left(\frac{1}{1 + \frac{1}{x}} \right)$$

$$= -\frac{1}{2} \cdot x^{-3/2} \cdot \frac{x}{x+1}$$

one more step to write it in regular form

Answer $6\sqrt{6} - 2\sqrt{3}$

b) $\int_1^{18} \sqrt{\frac{3}{x}} dx$ (1)

$$= \sqrt{3} \int_1^{18} \frac{1}{\sqrt{x}} dx$$

$$= \sqrt{3} \left(2x^{1/2} \right) \Big|_1^{18}$$

$$= \sqrt{3} (2\sqrt{18} - 2\sqrt{1})$$

$$= 2\sqrt{3} (2\sqrt{18} - 2)$$

$$= 2\sqrt{3} (2\sqrt{18} - 2)$$

Discontinuous at $x=0$

for $f(0)$ with limit right limit

$$\int \left(4 + 4(2) - \frac{1}{3}(8) \right) = \frac{28}{3}$$

$$\int \left(\frac{x^2}{2} + 2x \right)$$

$$f(x) = (x+2)$$

$$f(x) = -(x+2)$$

$$\int_{-1}^3 f(x) dx = 12$$

Answer is 4.

(2 marks)

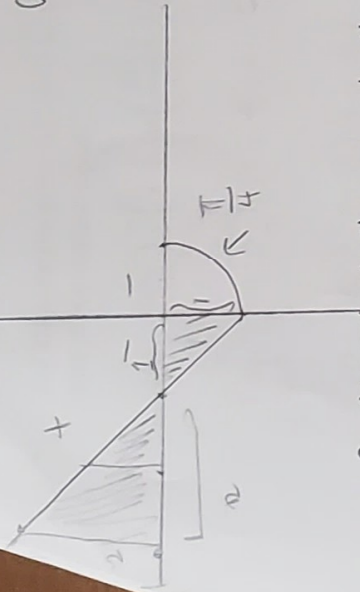
Application (10 marks)

- Graph function f given by: $f(x) = \begin{cases} -x-1 & \text{if } -3 \leq x \leq 0 \\ -\sqrt{1-x^2} & \text{if } 0 \leq x \leq 1 \end{cases}$. Using areas, calculate the integral $\int_{-3}^1 f(x) dx$

$$\int_{-3}^1 f(x) dx$$

To find area we use formula for area of a

3.5 / (4 marks)



Circle πr^2 side $r=1$ then

$$\pi \cdot \frac{1}{4} = \text{Area}$$

Area of triangle is

$$\left(\frac{2 \cdot 2}{2} \right) \text{ and } \frac{1 \cdot 1}{2}$$

Total area

$$2 - \frac{1}{2} \cdot \frac{\pi}{4}$$

$$\frac{3}{2} \pm \frac{\pi}{4}$$

$$\frac{6\pi}{4} - \frac{1}{4} = \frac{6\pi - 1}{4}$$

$$\text{Ans: } \frac{\pi}{2} - \frac{1}{4}$$

- Approximate the area under the curve $y = x^3 + 4x$ from $x=0$ to $x=8$ using $n=4$: (2 marks)

a) LRAM

$$\Delta x = 2$$

Using LRAM

$$\Delta x (f(0) + f(2) + f(4) + f(6))$$

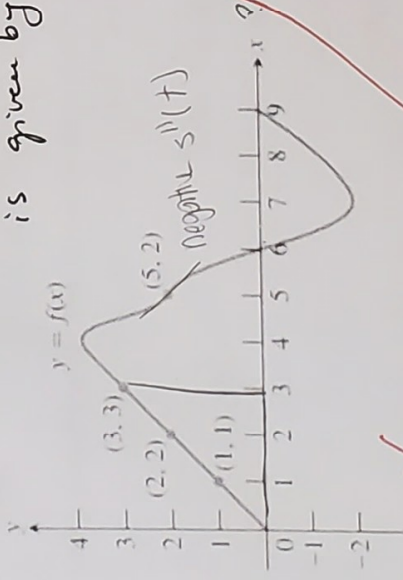
$$\Delta x (0 + 16 + 80 + 240)$$

$$1216$$

672 is also using LRAM

- f is differentiable function, graph is below. The position at time t (sec) is given by $s(t) = \int_0^t f(x) dx$

(4 marks)



a) 2 units/s $\frac{9}{2}$ units

b) Negative

Be $s'(t)$ is negative

position is

$$s(3) = 4.5$$

Communication (4 marks)

- Which RAM is biggest: If a positive, continuous function is decreasing on an interval what can we say about the relative sizes of LRAM, MRAM and RRAM? Explain. (1 mark)

RRAM > LRAM

LRAM > MRAM > RRAM

heights decrease